BioSync KAN-Elite v3.0 (OPTIMIZED)

Advanced Alzheimer's Prediction System

Diagnosis & Longitudinal Forecasting: Present → 1Y → 2Y → 3Y → 4Y → 5Y

- GNN (Graph Attention Network)
- Transformer (Multi-head + KAN)
- LSTM (Bidirectional + Attention)
- XGBoost (Gradient Boosting)
- BioSync Fusion ★ (All combined)

Feature Engineering: Temporal velocity · Polynomial · RobustScaling

Optimizations: Mixed Precision · Gradient Accumulation · Memory Management

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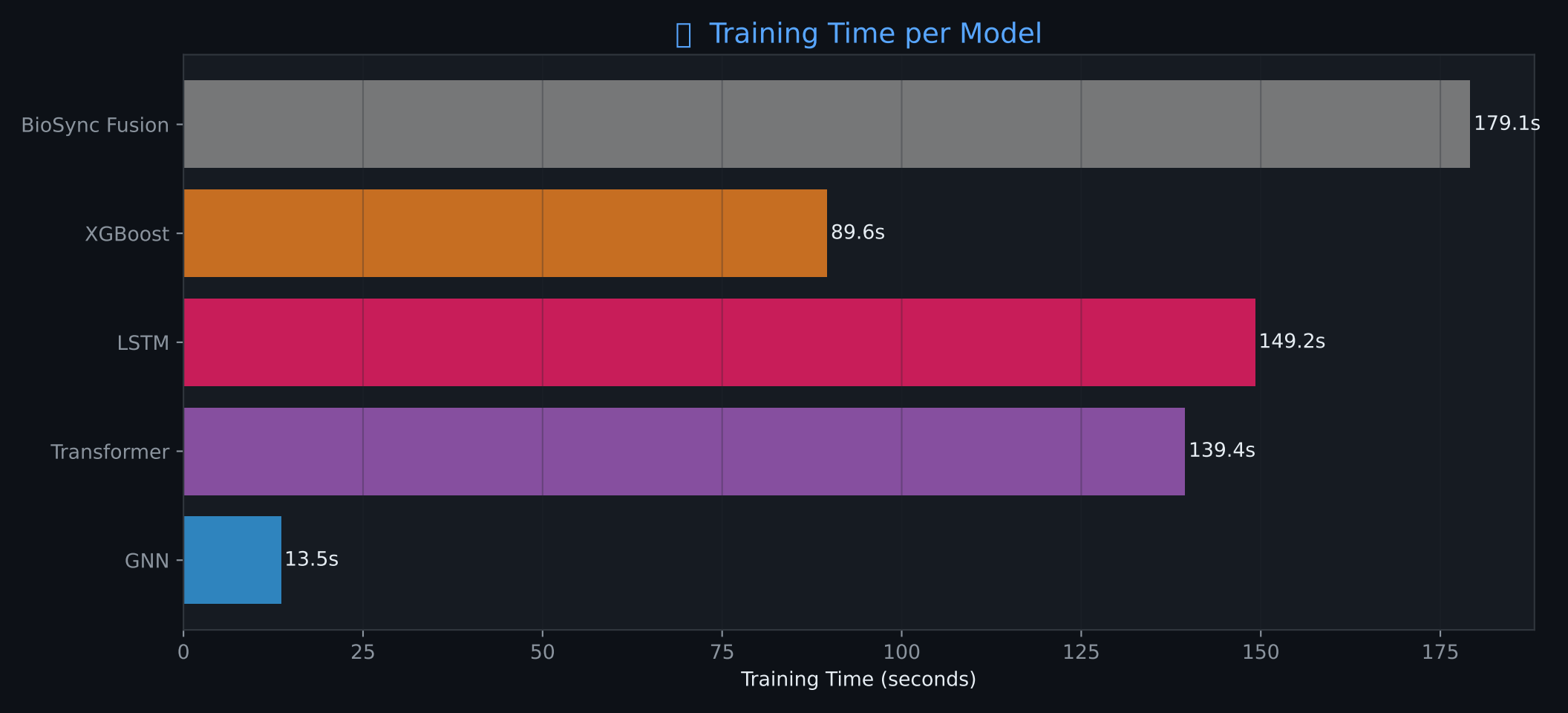
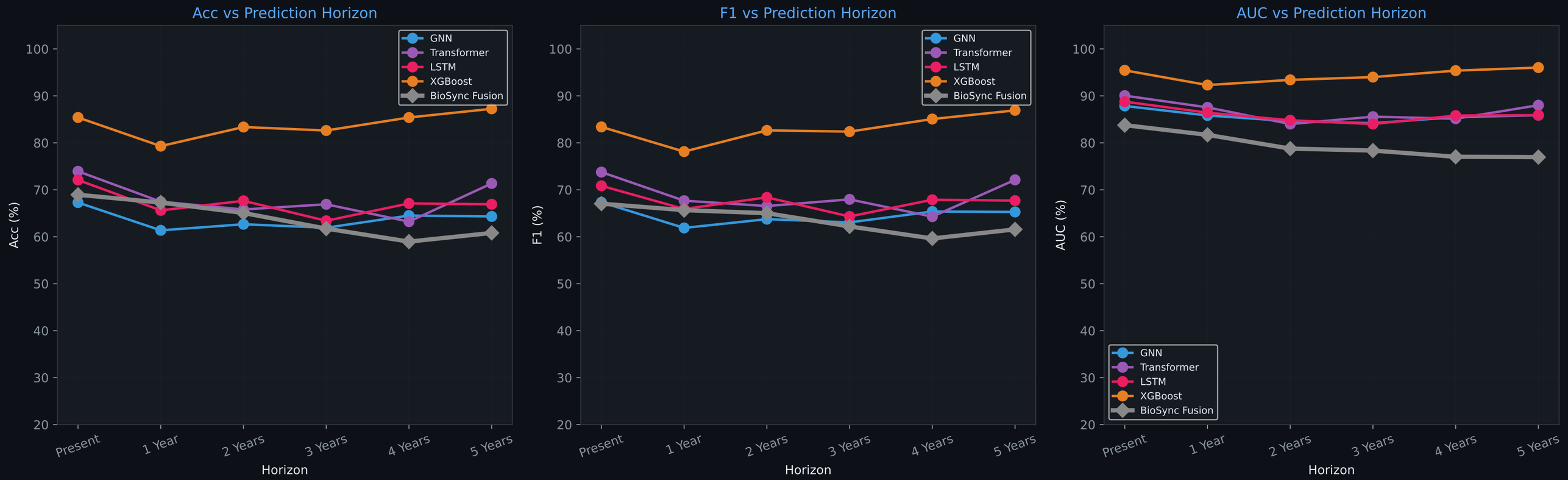

Training Time per Model
BioSync Fusion
XGBoost
LSTM
Transformer
GNN
179.1s
89.6s
149.2s
139.4s
13.5s
0
25
50
75
100
125
150
175
Training Time (seconds)

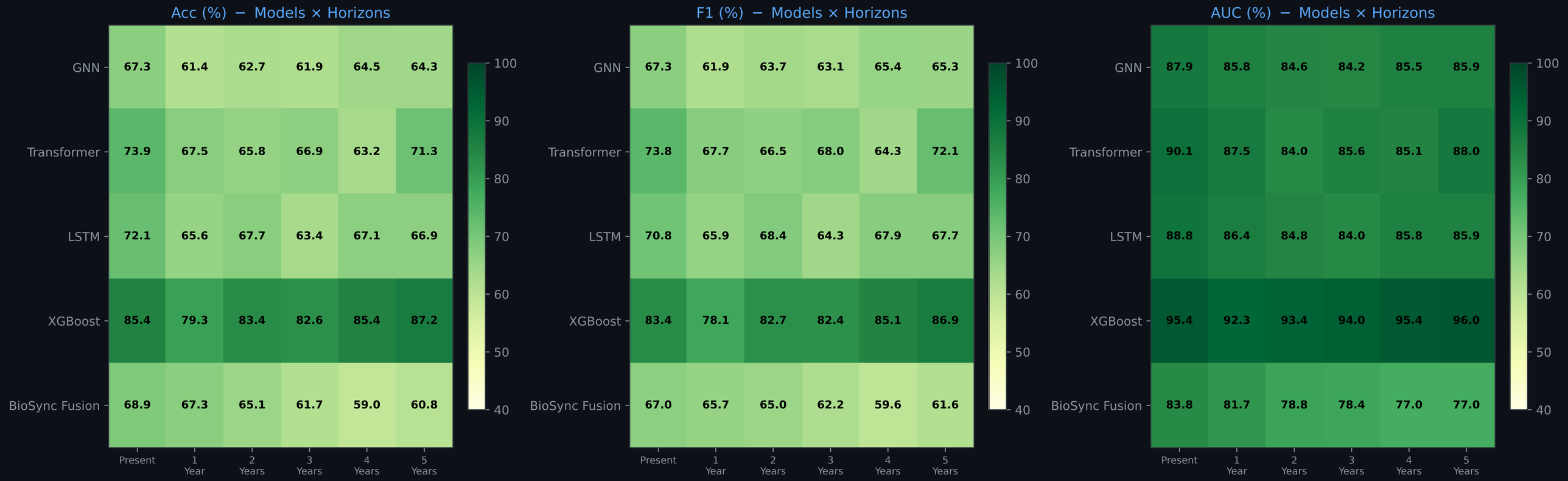
BioSync KAN-Elite — Learning Curves
Train
Val
Val (smooth)
Best epoch 89
Loss
7.0
6.5
6.0
5.5
5.0
4.5
4.0
3.5
0
20
40
60
80
100
120
Epoch

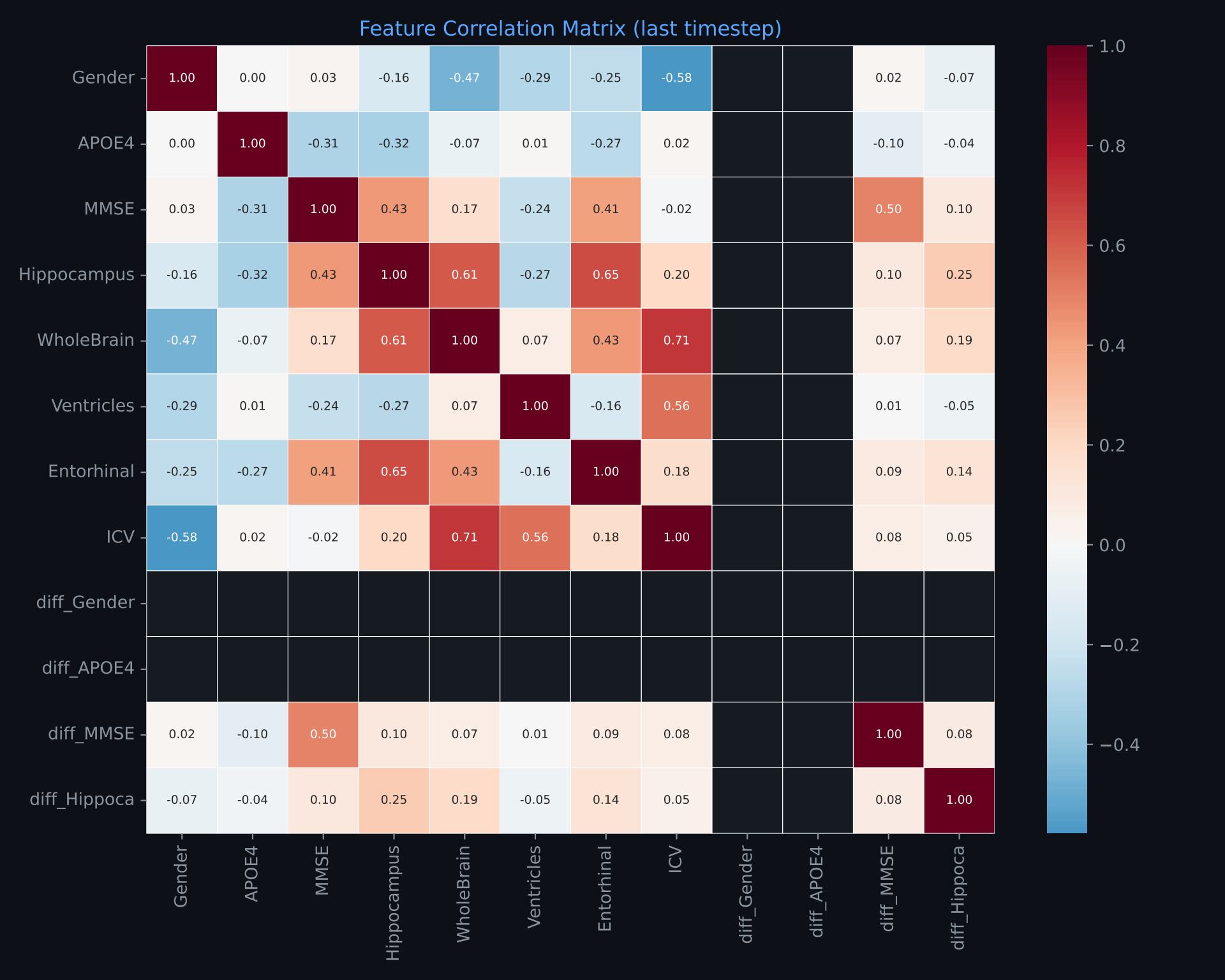
Model Comparison: Accuracy Across All Horizons

Performance Trend Across Horizons — All Models

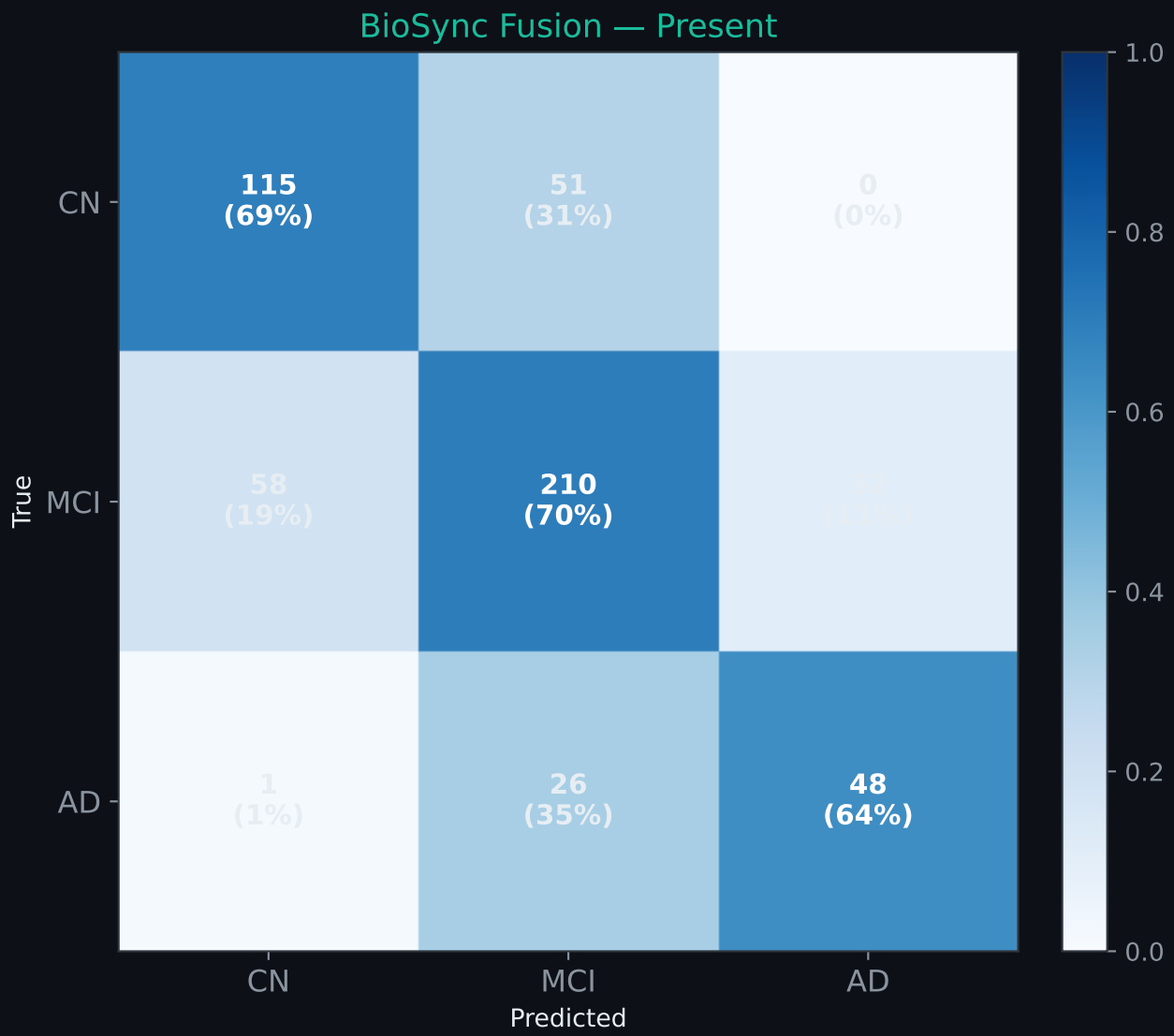


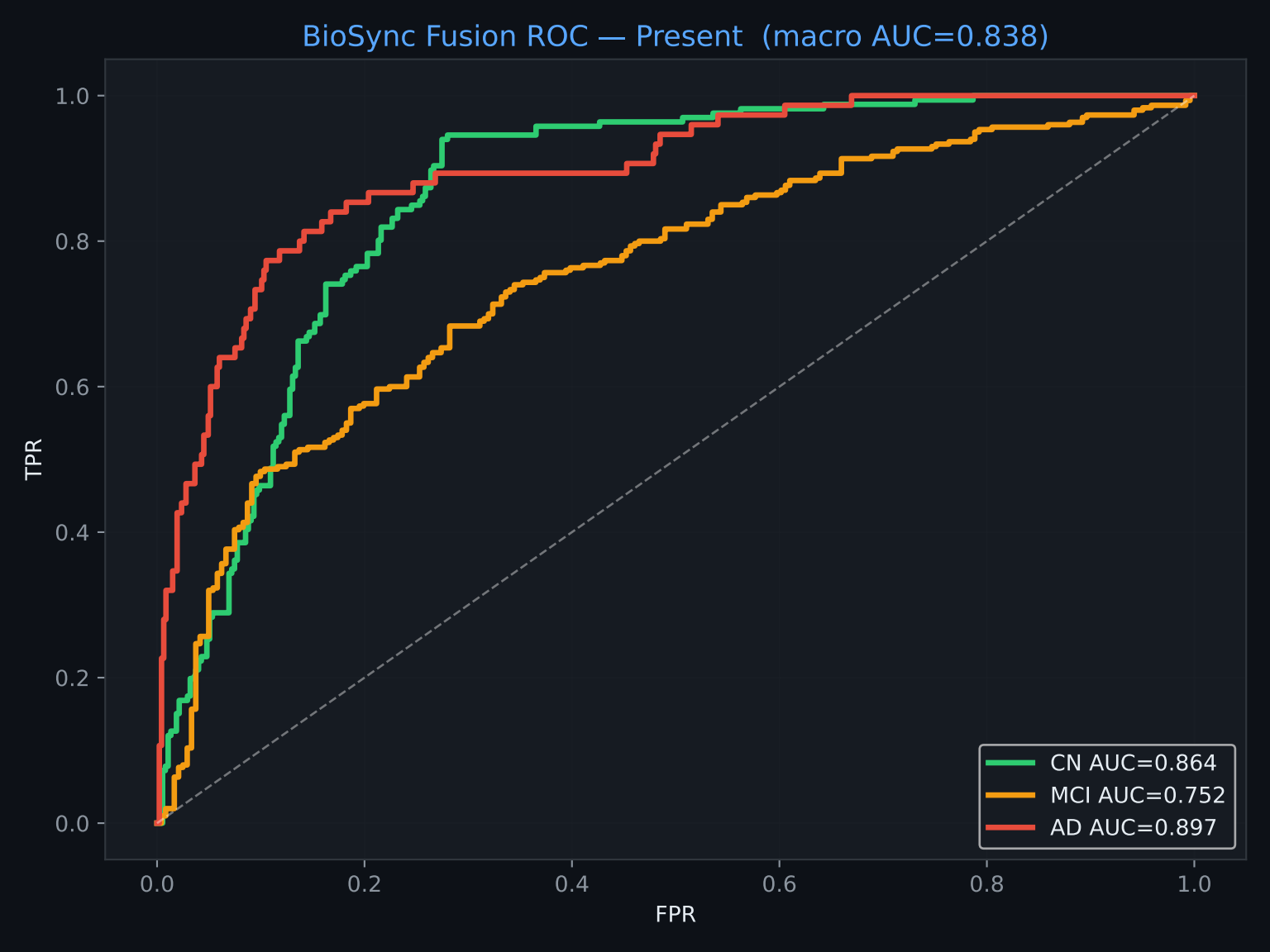
Comprehensive Performance Heatmap

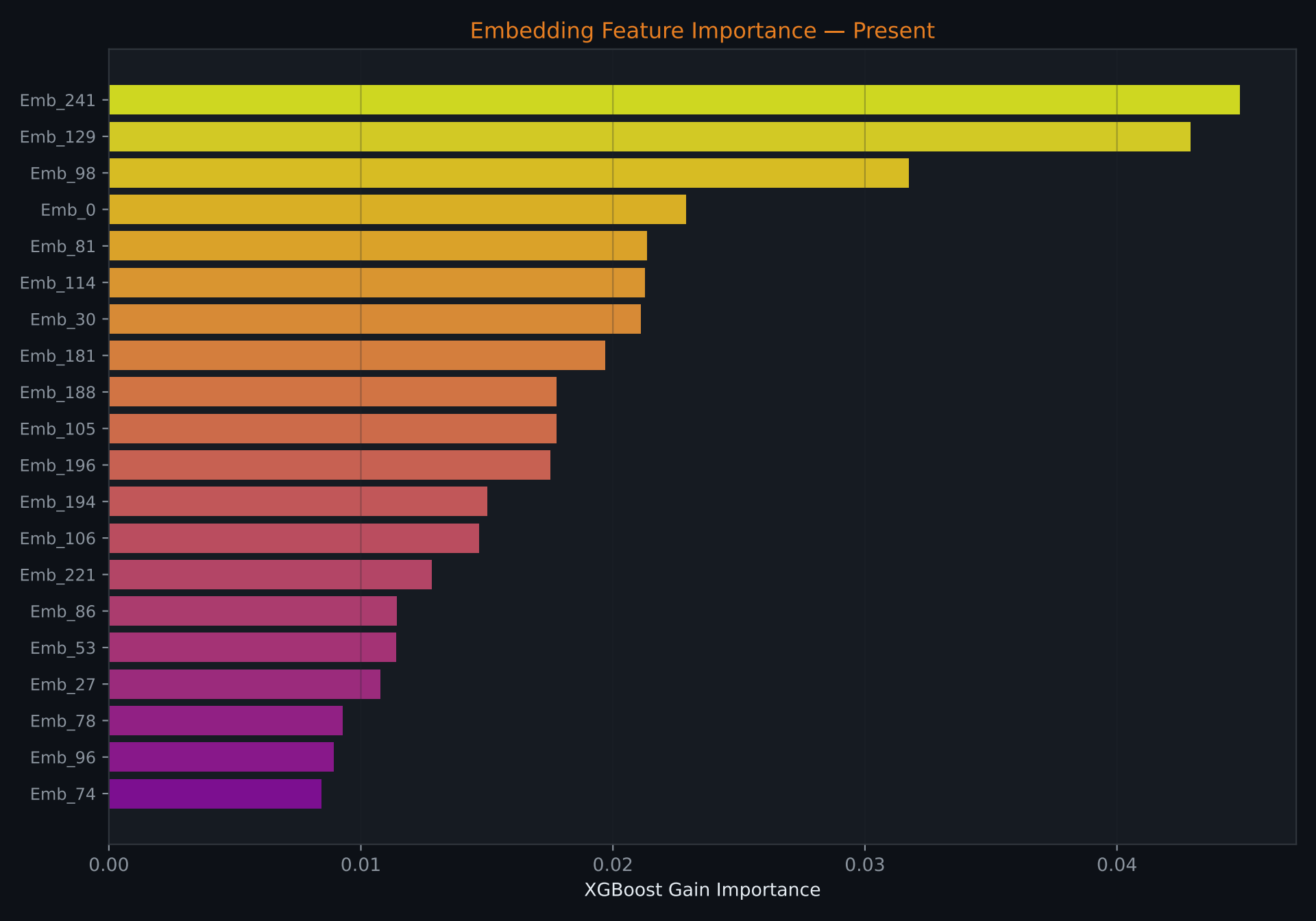


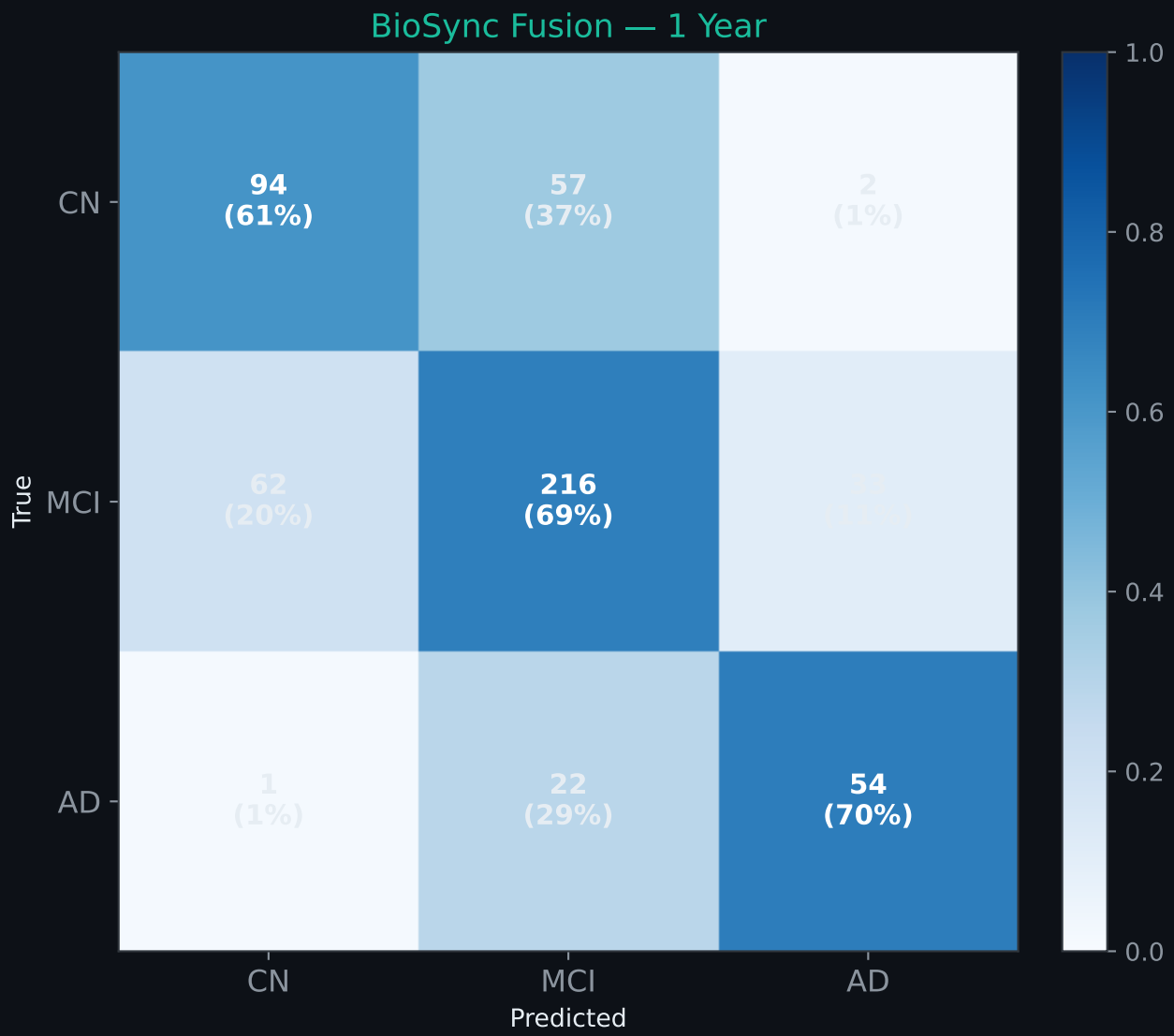

Feature Correlation Matrix (last timestep)
Gender 1.00 0.00 0.03 -0.16 -0.47 -0.29 -0.25 -0.58 0.02 -0.07
APOE4 0.00 1.00 -0.31 -0.32 -0.07 0.01 -0.27 0.02 -0.10 -0.04
MMSE 0.03 -0.31 1.00 0.43 0.17 -0.24 0.41 -0.02 0.50 0.10
Hippocampus -0.16 -0.32 0.43 1.00 0.61 -0.27 0.65 0.20 0.10 0.25
WholeBrain -0.47 -0.07 0.17 0.61 1.00 0.07 0.43 0.71 0.07 0.19
Ventricles -0.29 0.01 -0.24 -0.27 0.07 1.00 -0.16 0.56 0.01 -0.05
Entorhinal -0.25 -0.27 0.41 0.65 0.43 -0.16 1.00 0.18 0.09 0.14
ICV -0.58 0.02 -0.02 0.20 0.71 0.56 0.18 1.00 0.08 0.05
diff_Gender
diff_APOE4
diff_MMSE 0.02 -0.10 0.50 0.10 0.07 0.01 0.09 0.08 1.00 0.08
diff_Hippoca -0.07 -0.04 0.10 0.25 0.19 -0.05 0.14 0.05 0.08 1.00
Gender APOE4 MMSE Hippocampus WholeBrain Ventricles Entorhinal ICV diff_Gender diff_APOE4 diff_MMSE diff_Hippoca
1.0 0.8 0.6 0.4 0.2 0.0 −0.2 −0.4

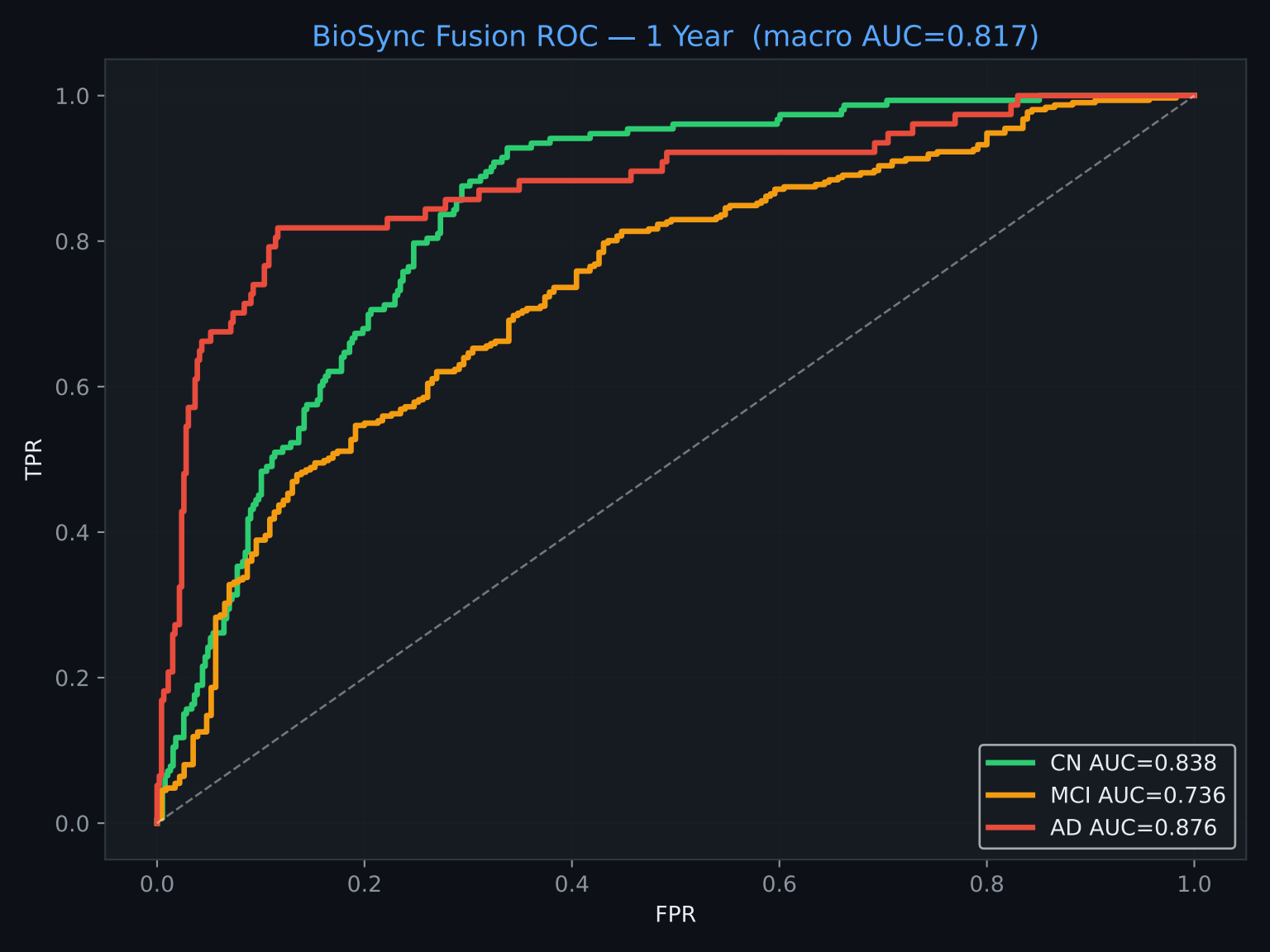
Feature Distribution by Diagnosis Class (CN / MCI / AD)

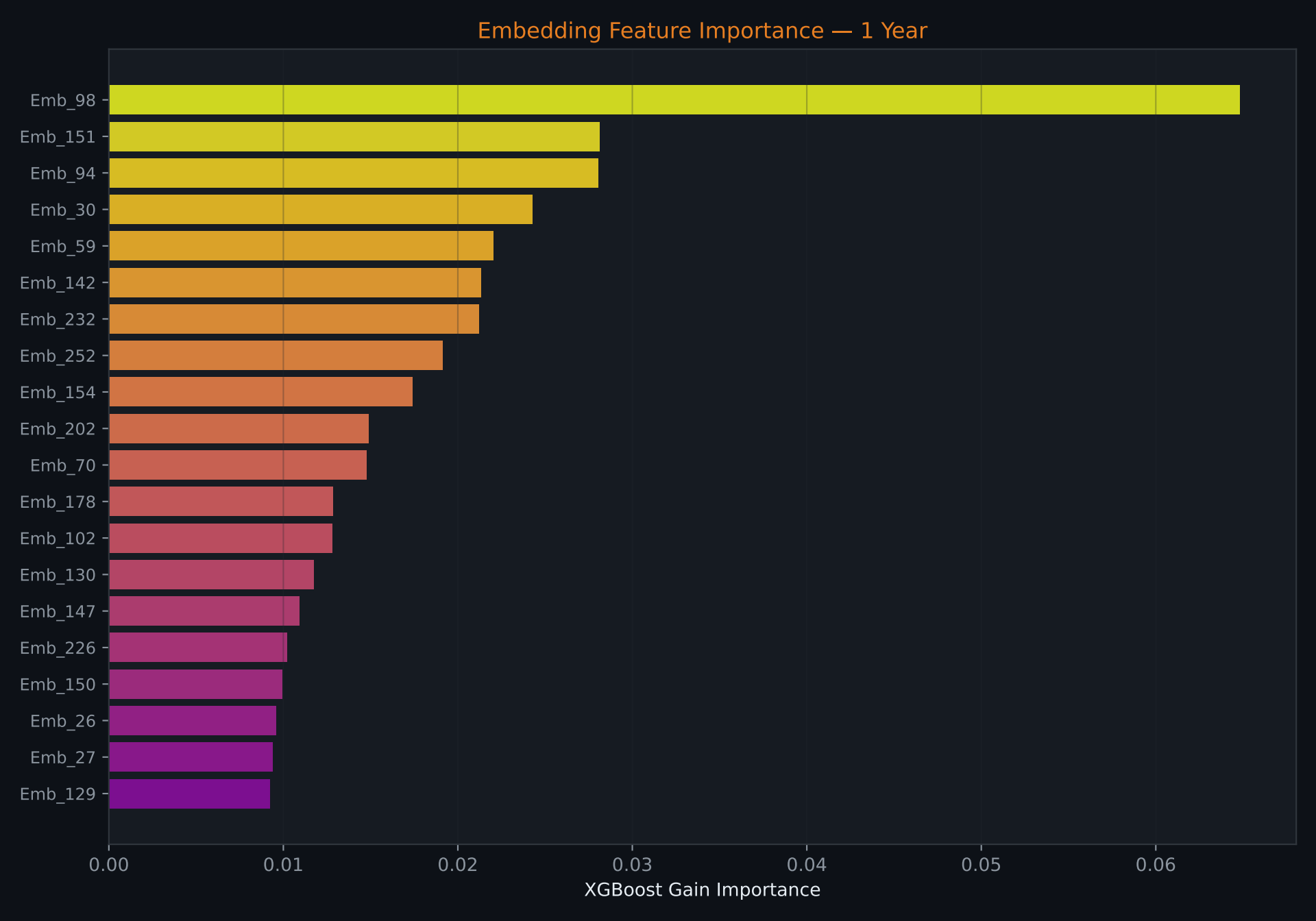

BioSync Fusion — Present
115
(69%)
51
(31%)
0
(0%)
58
(19%)
210
(70%)
32
(11%)
1
(1%)
26
(35%)
48
(64%)
CN
MCI
AD
True
CN
MCI
AD
Predicted
1.0
0.8
0.6
0.4
0.2
0.0

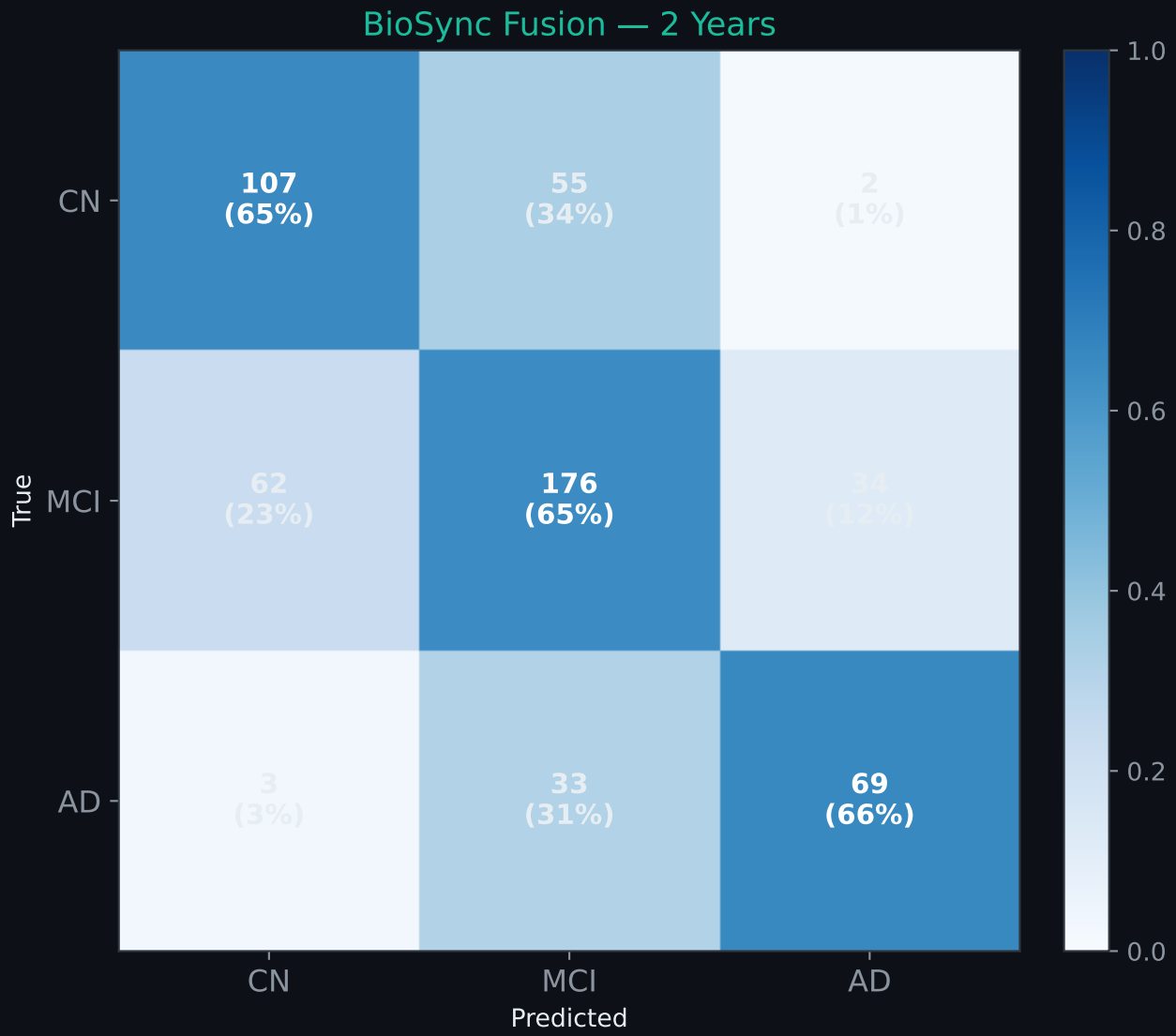

BioSync Fusion ROC — Present (macro AUC=0.838)
TPR
FPR
0.0
0.2
0.4
0.6
0.8
1.0
CN AUC=0.864
MCI AUC=0.752
AD AUC=0.897

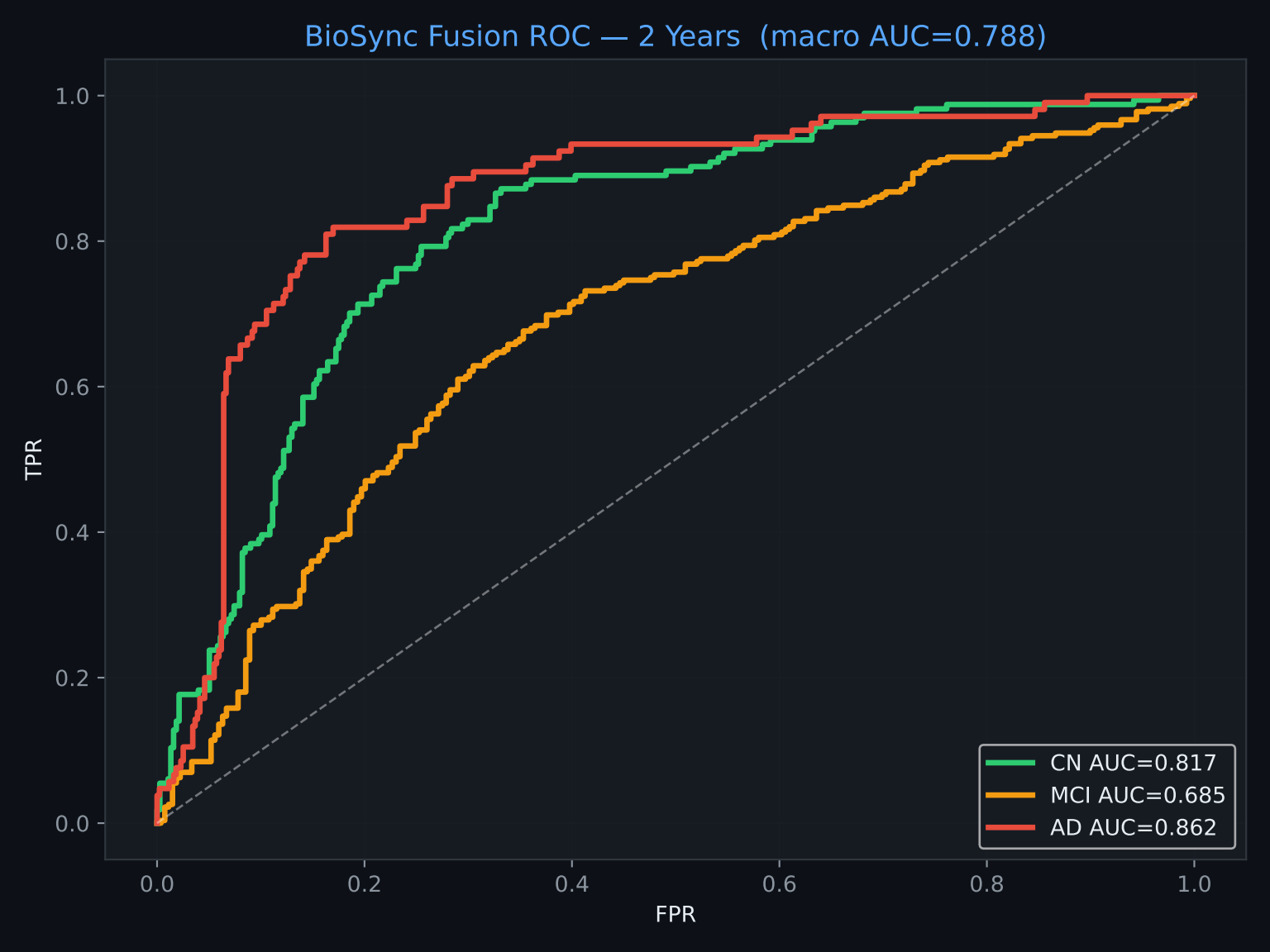

Embedding Feature Importance — Present
Emb_241
Emb_129
Emb_98
Emb_0
Emb_81
Emb_114
Emb_30
Emb_181
Emb_188
Emb_105
Emb_196
Emb_194
Emb_106
Emb_221
Emb_86
Emb_53
Emb_27
Emb_78
Emb_96
Emb_74
0.00
0.01
0.02
0.03
0.04
XGBoost Gain Importance

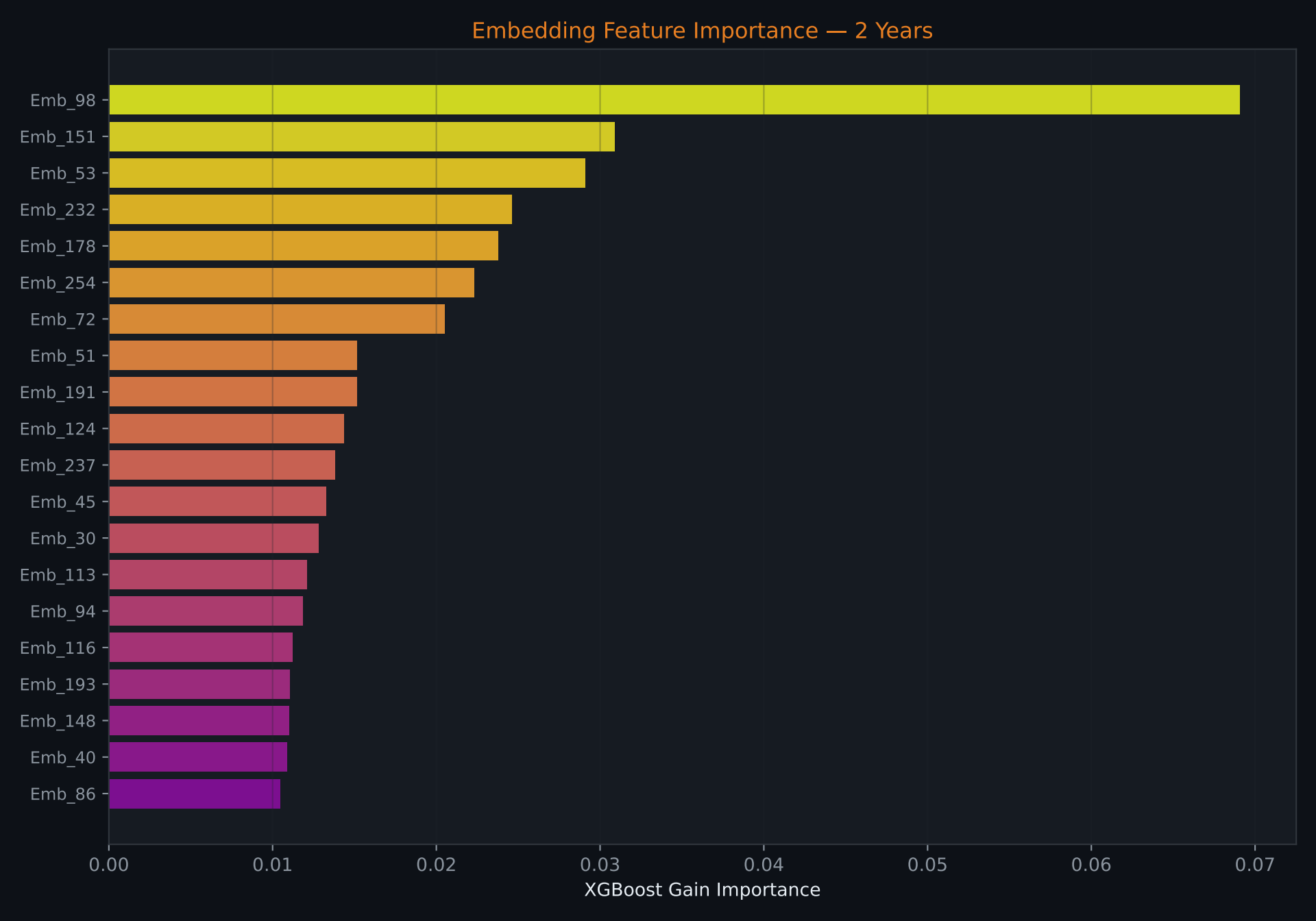

BioSync Fusion — 1 Year
True
CN
MCI
AD
94
(61%)
57
(37%)
2
(1%)
62
(20%)
216
(69%)
33
(11%)
1
(1%)
22
(29%)
54
(70%)
CN
MCI
AD
Predicted
1.0
0.8
0.6
0.4
0.2
0.0

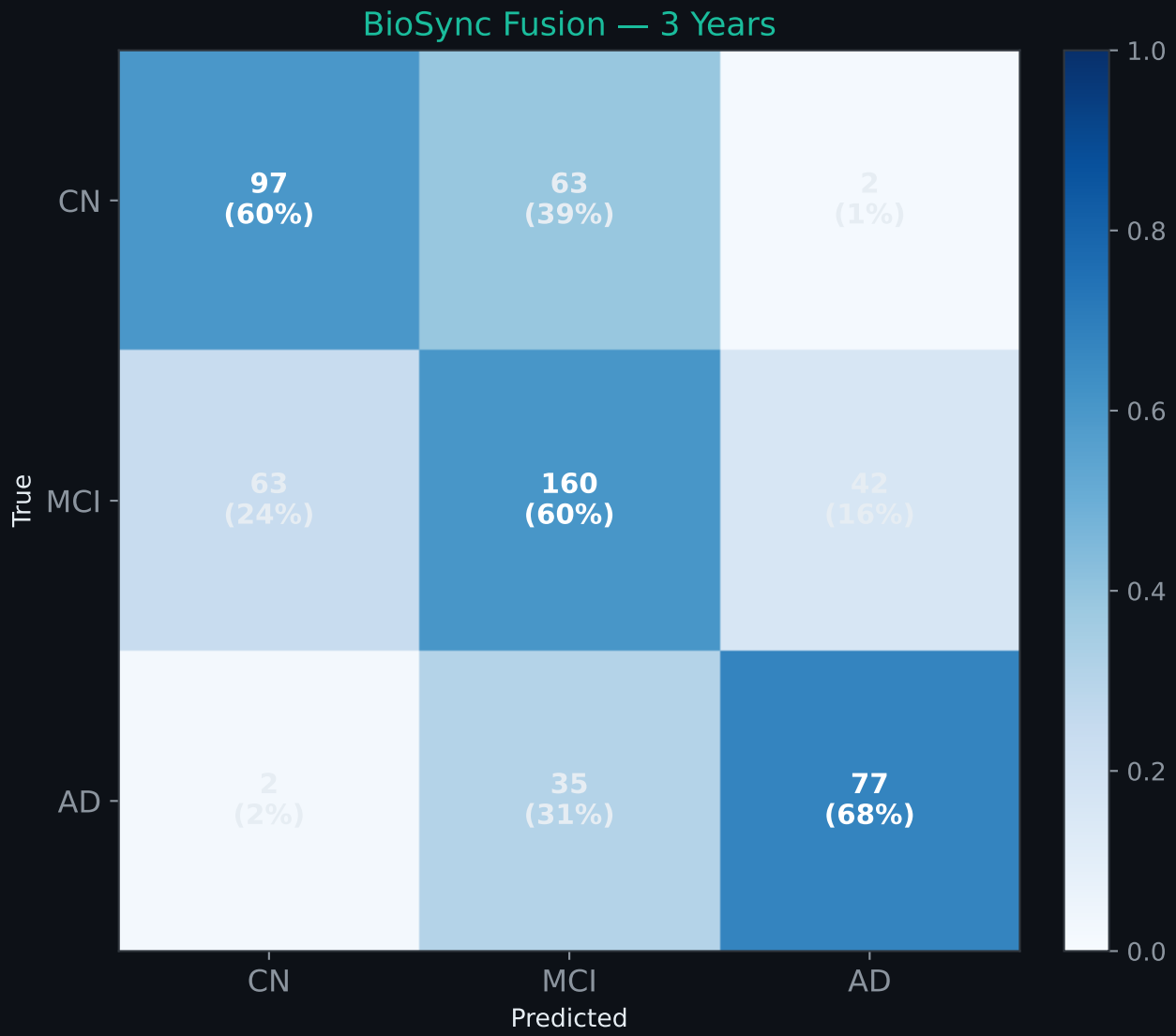

BioSync Fusion ROC — 1 Year (macro AUC=0.817)
TPR
FPR
0.0
0.2
0.4
0.6
0.8
1.0
CN AUC=0.838
MCI AUC=0.736
AD AUC=0.876

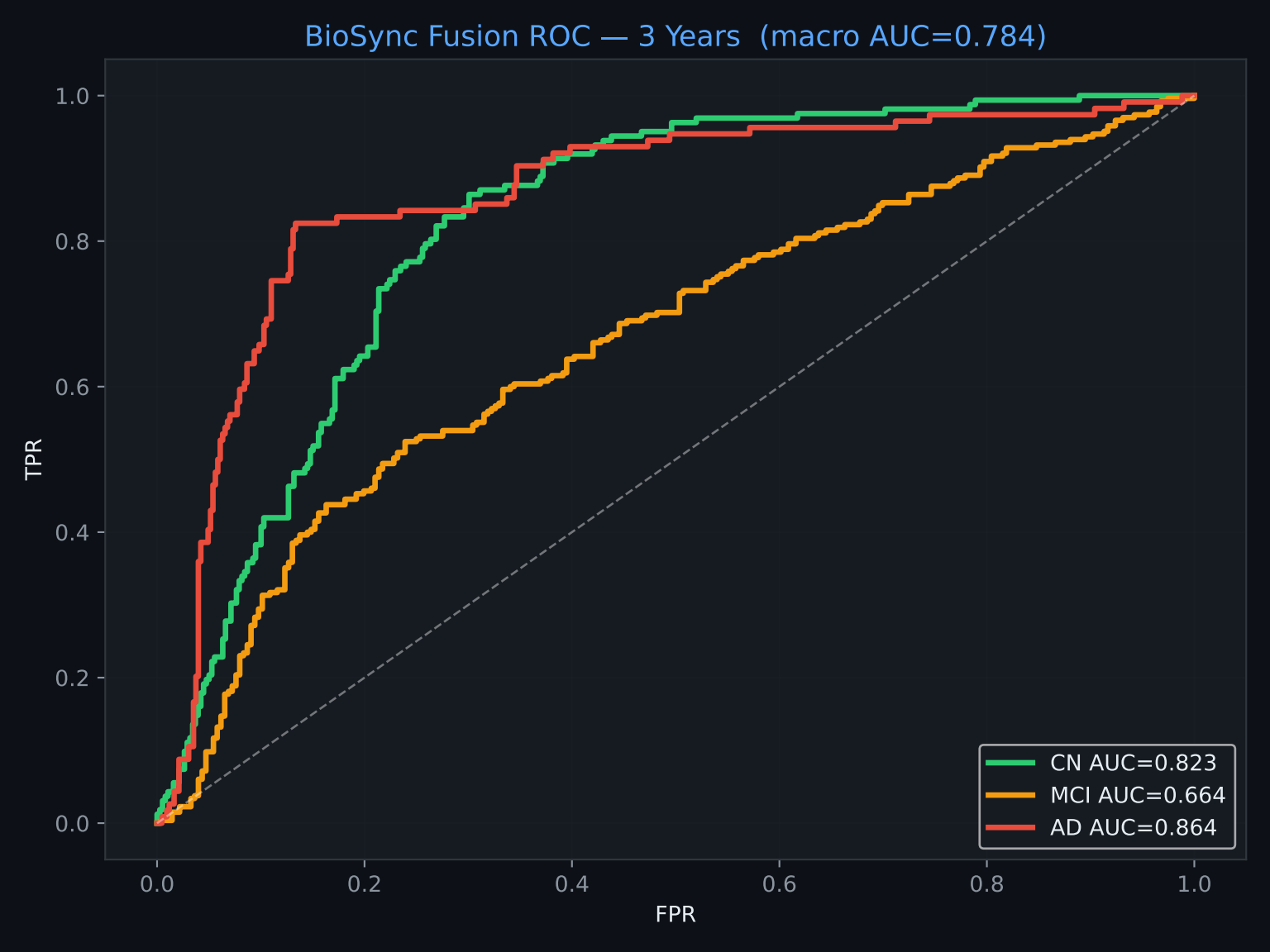

Embedding Feature Importance — 1 Year
Emb_98
Emb_151
Emb_94
Emb_30
Emb_59
Emb_142
Emb_232
Emb_252
Emb_154
Emb_202
Emb_70
Emb_178
Emb_102
Emb_130
Emb_147
Emb_226
Emb_150
Emb_26
Emb_27
Emb_129
0.00
0.01
0.02
0.03
0.04
0.05
0.06
XGBoost Gain Importance

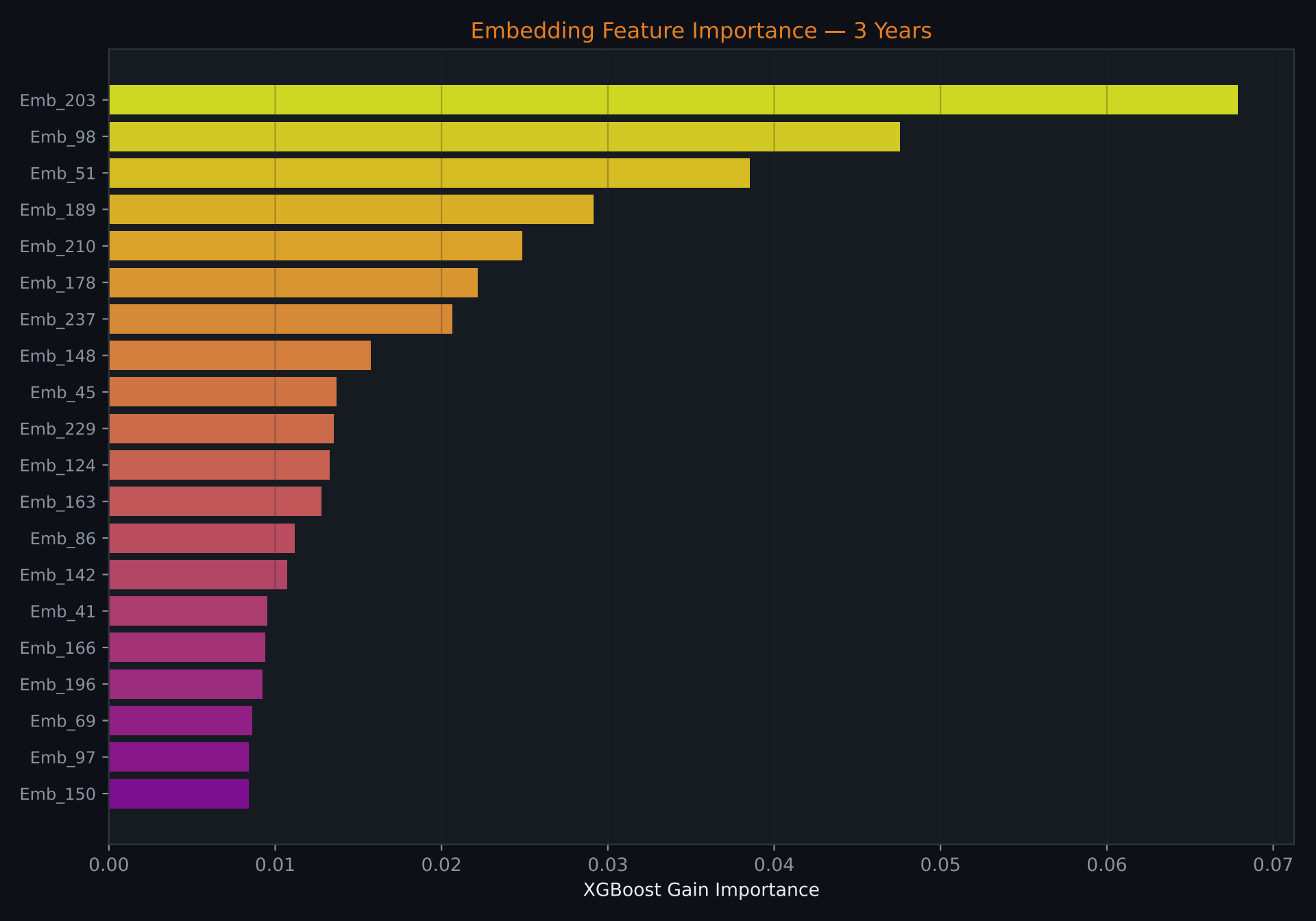

BioSync Fusion — 2 Years
True
CN
MCI
AD
107
(65%)
55
(34%)
2
(1%)
62
(23%)
176
(65%)
34
(12%)
3
(3%)
33
(31%)
69
(66%)
CN
MCI
AD
Predicted
1.0
0.8
0.6
0.4
0.2
0.0

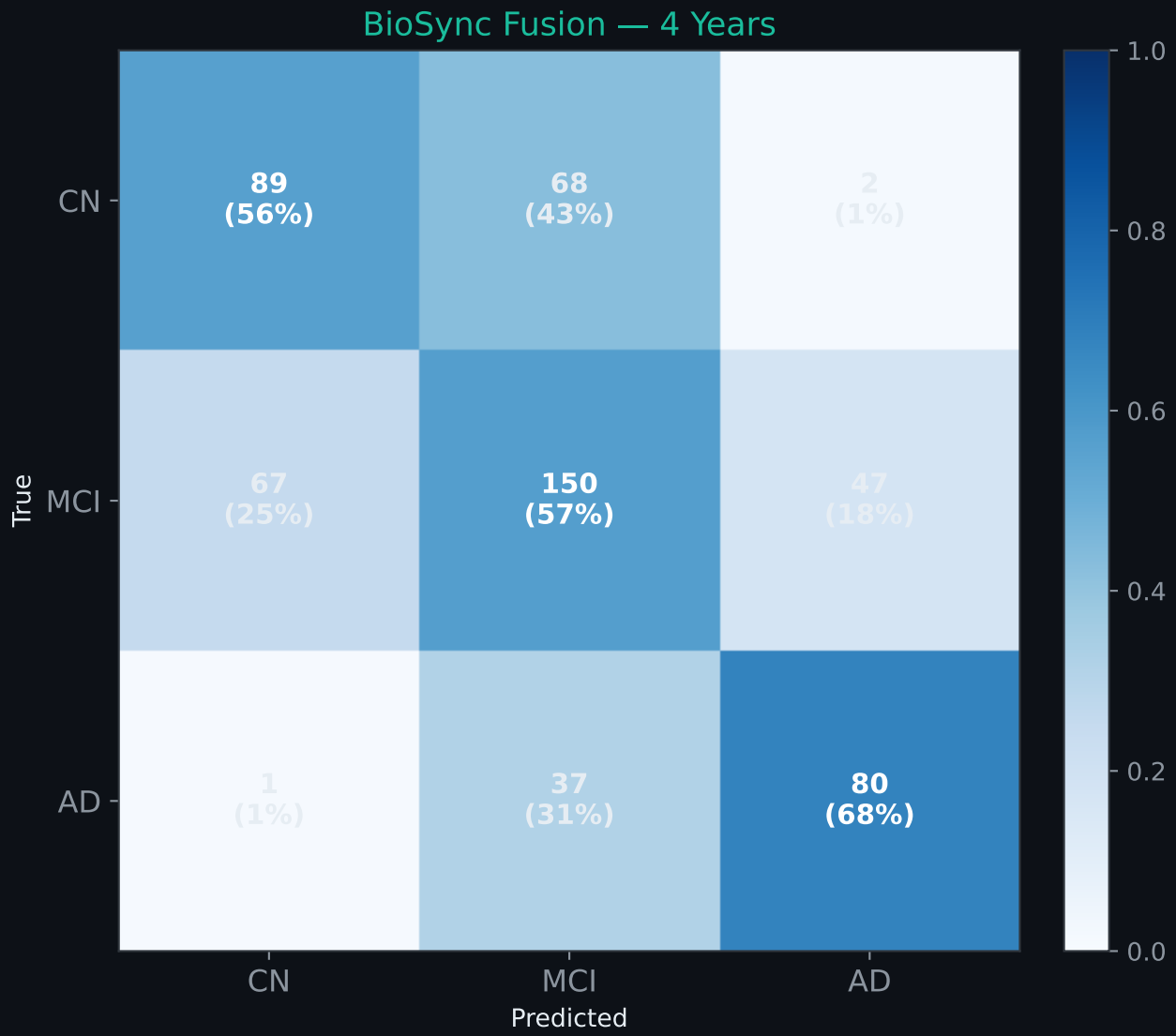

BioSync Fusion ROC — 2 Years (macro AUC=0.788)
TPR
FPR
0.0
0.2
0.4
0.6
0.8
1.0
CN AUC=0.817
MCI AUC=0.685
AD AUC=0.862

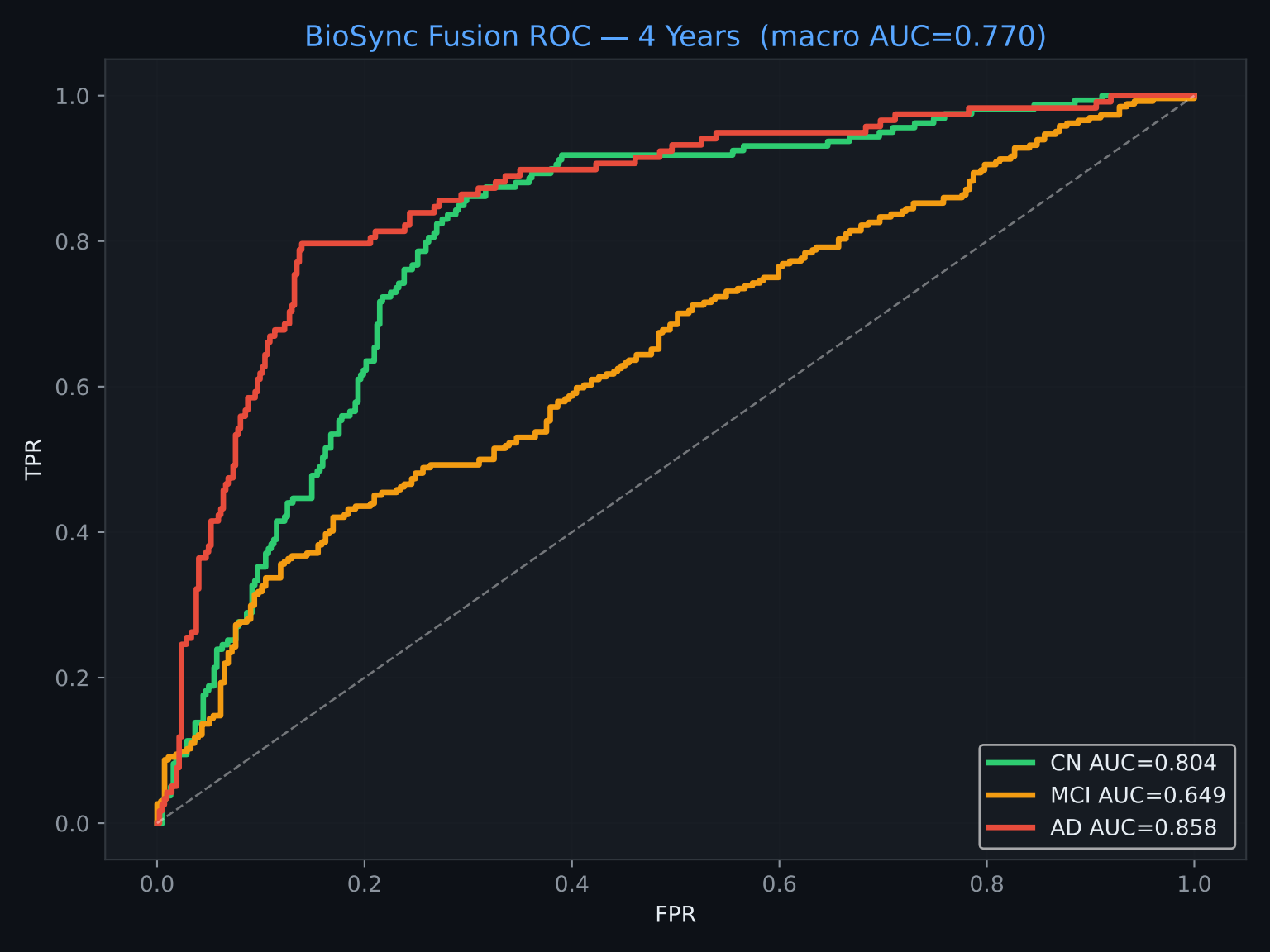

Embedding Feature Importance — 2 Years
Emb_98
Emb_151
Emb_53
Emb_232
Emb_178
Emb_254
Emb_72
Emb_51
Emb_191
Emb_124
Emb_237
Emb_45
Emb_30
Emb_113
Emb_94
Emb_116
Emb_193
Emb_148
Emb_40
Emb_86
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
XGBoost Gain Importance

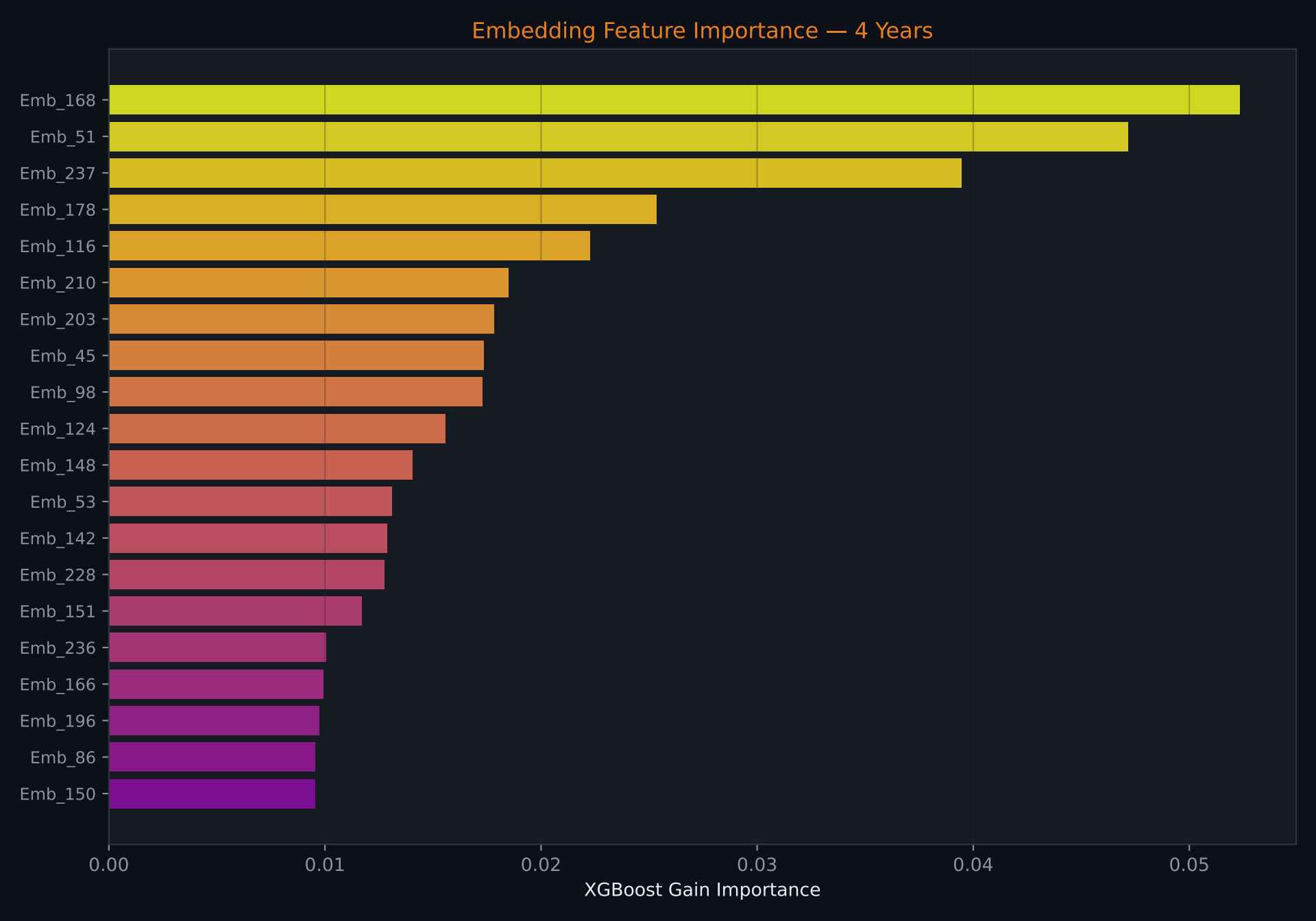

BioSync Fusion — 3 Years
True
CN
MCI
AD
97
(60%)
63
(39%)
2
(1%)
63
(24%)
160
(60%)
42
(16%)
2
(2%)
35
(31%)
77
(68%)
CN
MCI
AD
Predicted
1.0
0.8
0.6
0.4
0.2
0.0

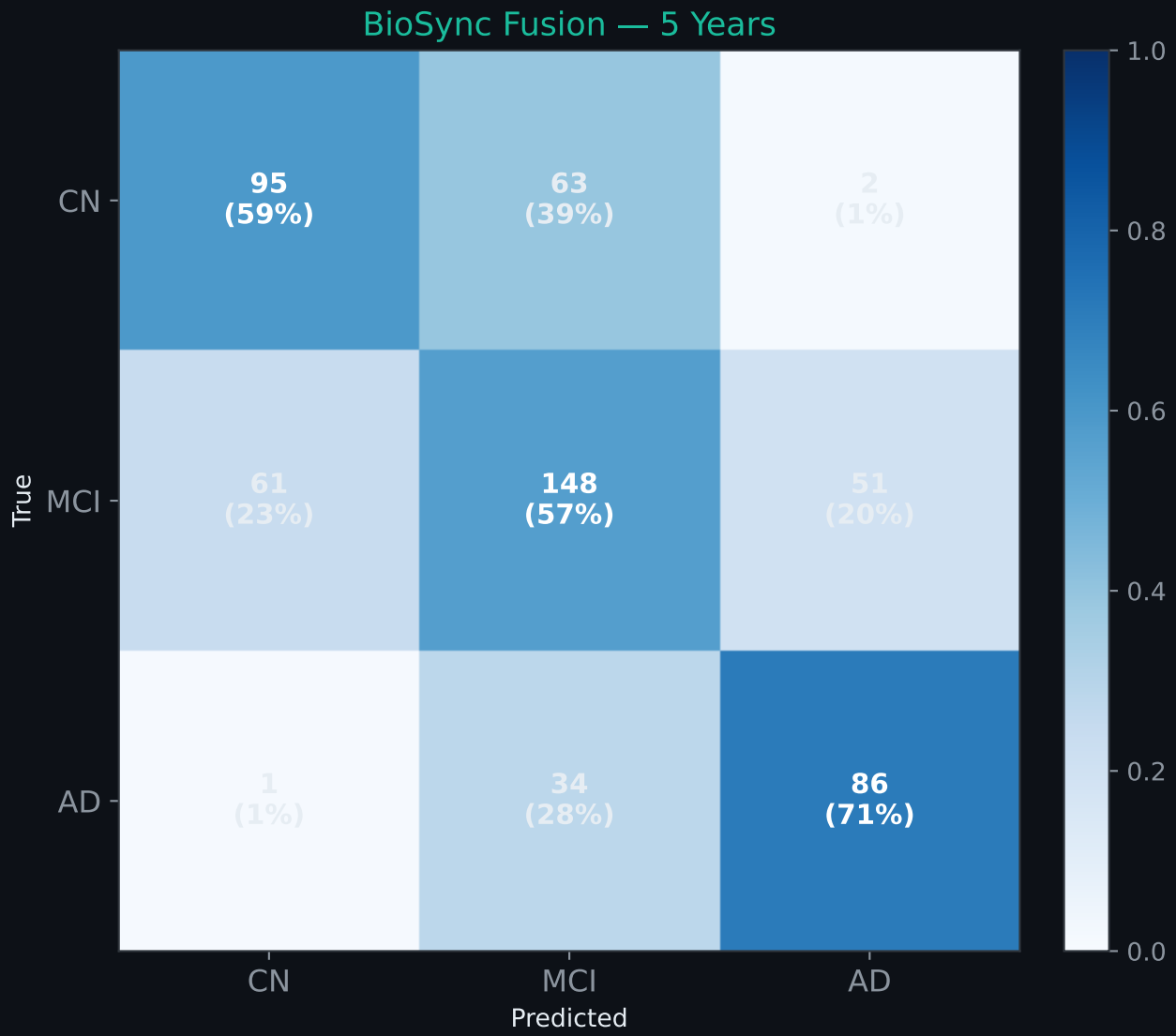

BioSync Fusion ROC — 3 Years (macro AUC=0.784)
TPR
FPR
0.0
0.2
0.4
0.6
0.8
1.0
CN AUC=0.823
MCI AUC=0.664
AD AUC=0.864

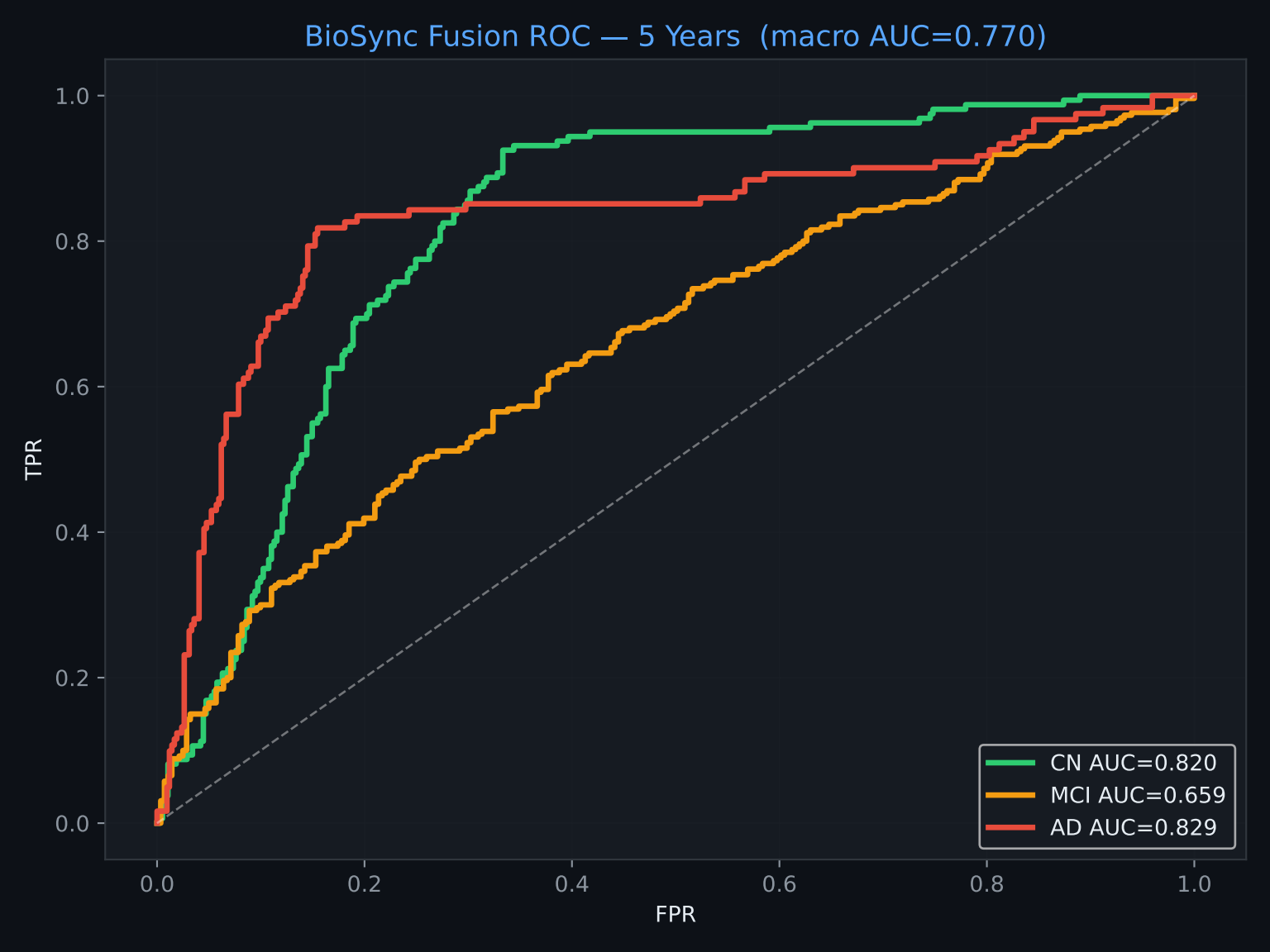

Embedding Feature Importance — 3 Years
Emb_203
Emb_98
Emb_51
Emb_189
Emb_210
Emb_178
Emb_237
Emb_148
Emb_45
Emb_229
Emb_124
Emb_163
Emb_86
Emb_142
Emb_41
Emb_166
Emb_196
Emb_69
Emb_97
Emb_150
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
XGBoost Gain Importance

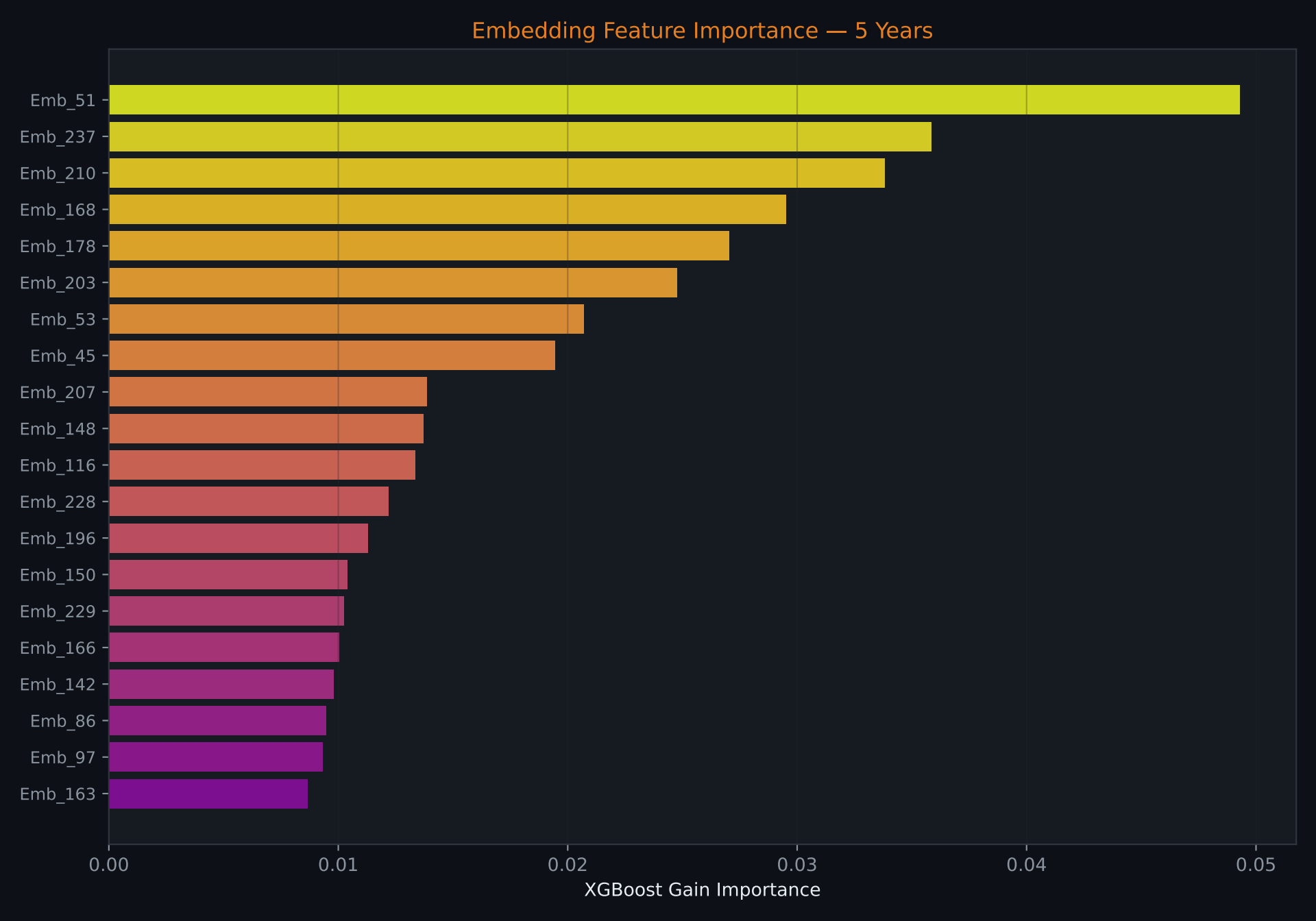

BioSync Fusion — 4 Years
True
Predicted
CN
MCI
AD
CN
MCI
AD
89
(56%)
68
(43%)
2
(1%)
67
(25%)
150
(57%)
47
(18%)
1
(1%)
37
(31%)
80
(68%)
1.0
0.8
0.6
0.4
0.2
0.0


BioSync Fusion ROC — 4 Years (macro AUC=0.770)
TPR
FPR
0.0
0.2
0.4
0.6
0.8
1.0
CN AUC=0.804
MCI AUC=0.649
AD AUC=0.858


Embedding Feature Importance — 4 Years
Emb_168
Emb_51
Emb_237
Emb_178
Emb_116
Emb_210
Emb_203
Emb_45
Emb_98
Emb_124
Emb_148
Emb_53
Emb_142
Emb_228
Emb_151
Emb_236
Emb_166
Emb_196
Emb_86
Emb_150
0.00
0.01
0.02
0.03
0.04
0.05
XGBoost Gain Importance


BioSync Fusion — 5 Years
True
Predicted
CN
MCI
AD
95
(59%)
63
(39%)
2
(1%)
61
(23%)
148
(57%)
51
(20%)
1
(1%)
34
(28%)
86
(71%)
1.0
0.8
0.6
0.4
0.2
0.0


BioSync Fusion ROC — 5 Years (macro AUC=0.770)
TPR
FPR
0.0
0.2
0.4
0.6
0.8
1.0
CN AUC=0.820
MCI AUC=0.659
AD AUC=0.829


Embedding Feature Importance — 5 Years
Emb_51
Emb_237
Emb_210
Emb_168
Emb_178
Emb_203
Emb_53
Emb_45
Emb_207
Emb_148
Emb_116
Emb_228
Emb_196
Emb_150
Emb_229
Emb_166
Emb_142
Emb_86
Emb_97
Emb_163
0.00
0.01
0.02
0.03
0.04
0.05
XGBoost Gain Importance

Complete Performance Summary — All Models × All Horizons

Model	Horizon	Acc%	F1%	AUC%	Prec%
GNN	Present	67.3	67.3	87.9	66.1
GNN	1 Year	61.4	61.9	85.8	61.4
GNN	2 Years	62.7	63.7	84.6	62.9
GNN	3 Years	61.9	63.1	84.2	62.0
GNN	4 Years	64.5	65.4	85.5	64.5
GNN	5 Years	64.3	65.3	85.9	64.3
Transformer	Present	73.9	73.8	90.1	71.5
Transformer	1 Year	67.5	67.7	87.5	65.6
Transformer	2 Years	65.8	66.5	84.0	65.5
Transformer	3 Years	66.9	68.0	85.6	66.8
Transformer	4 Years	63.2	64.3	85.1	63.8
Transformer	5 Years	71.3	72.1	88.0	71.0
LSTM	Present	72.1	70.8	88.8	70.6
LSTM	1 Year	65.6	65.9	86.4	64.4
LSTM	2 Years	67.7	68.4	84.8	67.4
LSTM	3 Years	63.4	64.3	84.0	63.6
LSTM	4 Years	67.1	67.9	85.8	66.7
LSTM	5 Years	66.9	67.7	85.9	66.6
XGBoost	Present	85.4	83.4	95.4	84.5
XGBoost	1 Year	79.3	78.1	92.3	77.6
XGBoost	2 Years	83.4	82.7	93.4	82.5
XGBoost	3 Years	82.6	82.4	94.0	82.4
XGBoost	4 Years	85.4	85.1	95.4	85.2
XGBoost	5 Years	87.2	86.9	96.0	86.9
BioSync Fusion	Present	68.9	67.0	83.8	66.4
BioSync Fusion	1 Year	67.3	65.7	81.7	64.6
BioSync Fusion	2 Years	65.1	65.0	78.8	64.9
BioSync Fusion	3 Years	61.7	62.2	78.4	61.8
BioSync Fusion	4 Years	59.0	59.6	77.0	59.2
BioSync Fusion	5 Years	60.8	61.6	77.0	60.9

FINAL RESULTS — BioSync KAN-Elite v3.0 (OPTIMIZED)

GNN	Acc=63.7%	F1=64.4%	AUC=85.6%
Transformer	Acc=68.1%	F1=68.7%	AUC=86.7%
LSTM	Acc=67.1%	F1=67.5%	AUC=85.9%
XGBoost	Acc=83.9%	F1=83.1%	AUC=94.4%
★ **BioSync Fusion**	**Acc=63.8%**	**F1=63.5%**	**AUC=79.4%**

Training Time (seconds):

data	3.7s
GNN	13.5s
Transformer	139.4s
LSTM	149.2s
XGBoost	89.6s
BioSync Fusion	179.1s